

Rachel Williams

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U.S. Citizen
[Website](#)

EDUCATION

Cornell University, Ithaca NY — B.S. Mechanical Engineering

Expected May 2028

GPA: 3.87/4.00

Georgia Tech Europe, Metz France — Study Abroad Program

August 2026 - December 2026

WORK EXPERIENCE

Cornell University, Ithaca NY — Undergraduate Researcher

May 2026 - July 2026

- Optimized and standardized speckle patterns used in Digital Image Correlation (DIC) at the Cornell High Energy Synchrotron Source
- Developed quantitative speckle quality metrics evaluated with ImageJ
- Designed, executed, and documented experiments to analyze more than 30 different speckle pattern variations

Exeter Animal Shelter, Exeter RI — Volunteer

June 2025 - August 2025

Salk's Ace Hardware, North Kingstown RI — Cashier

June 2024 - August 2024

PROJECTS

Spotted Lanternfly Vineyard Mitigation Device (Class Design Project)

- Designed and fabricated a prototype lure-and-trap system to mitigate invasive spotted lanternflies in vineyards
- Collaborated with a five-member team to develop design concepts, evaluate engineering tradeoffs, and present the final design in a poster session

Woodworking Projects

- Built 16 custom picture frames using a hand held router
- Currently constructing an 18th-century flintlock rifle replica using traditional woodworking techniques

Rhode Island Science Olympiad (High School)

- Earned four gold medals in mechanically propelled vehicle and balsa wood structure challenges (2023-2024)
- Designed devices in CAD and fabricated them using laser cutting and a CNC router

TECHNICAL SKILLS

Programming

- Python
- MATLAB

CAD & Manufacturing

- Fusion 360
- 3D printing
- Laser cutting

Fabrication

- Manual lathe
- Manual mill
- Drill press
- Band saw
- Hand router
- Angle grinder
- MIG welder

Laboratory Tools

- ImageJ
- PulseRay RAMS IV load frame
- MEDM, EPICS, SPEC

LEADERSHIP & ACTIVITIES

President of the Cornell Cycling Club

- Established a bicycle loaner program to improve accessibility for new club members
- Organized and led a multi-day cycling training trip for club members

Clarinetist

- Studies with Dr. Richard Faria (Ithaca College)